

WORKY APP

Comprehensive Engineering & Project Management Baseline Standard Documentation

An all-in-one Progressive Web Application (PWA) designed for real-time discovery of study environments, automated seat & amenity verification, and location-based dynamic loyalty rewards.

ACADEMIC SUPERVISION & ASSESSMENT

Course BS213: Project Management Board | Academic Oversight Panel

PROJECT MANAGEMENT & CORE DEVELOPMENT TEAM

- Marwan Bahaa (Project Manager) — ID: 24030006
- Mohamed Gouda (Lead Developer) — ID: 24030145
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VERSION CONTROL & OPEN SOURCE REPOSITORY

GitHub Public Repository: Project2 Repository

KEY PARAMETERS

Baseline Schedule: 80 Days (8 Sprints) | Financial Budget Cap: 35,120 EGP | Document
Page Target: 20 Pages

TABLE OF CONTENTS

01. Executive Summary & Document Overview	Page 2
02. Project Charter & Governance Framework	Page 3
03. Feasibility Study (TELOS Framework Depth Analysis)	Page 4
04. Detailed Project Scope & System Architecture	Page 5
05. Work Breakdown Structure (WBS) & Dictionary	Page 6
06. Activity List & Dependency Schedule	Page 7
07. Project Cost Estimation Baseline & Bottom-Up Budgeting	Page 8
08. Earned Value Management (EVM) Deep Dive Baseline	Page 9
09. PERT Statistical Analysis & Schedule Risk Quantification	Page 10
10. Detailed Project Gantt Timeline (80-Day 8-Sprint Matrix)	Page 11
11. Critical Path Method (CPM) & Network Diagram Topology	Page 12
12. Comprehensive Risk Management & Contingency Matrix	Page 13
13. Technical Software Architecture & Stack Design	Page 14
14. Module Breakdown: Map Engine & Real-Time Geofencing	Page 15
15. Module Breakdown: QR Code Check-in & Loyalty Subsystem	Page 16
16. Module Breakdown: Partner Business Portal & Analytics	Page 17
17. Quality Assurance, Security & Regulatory Compliance	Page 18
18. Stakeholder Engagement & Communication Plan	Page 19
19. LMS Deliverables Verification & Project Sign-Off	Page 20

Executive Summary

The **WORKY Smart Co-Working Discovery PWA Platform** is an innovative mobile-first software solution developed specifically to bridge the communication and functional gap between university students/freelancers and modern co-working spaces in urban hubs. In academic and professional settings across Egypt, finding reliable, quiet, and well-equipped workspaces equipped with verified high-speed Wi-Fi, operational power sockets, and noise-managed study zones is a persistent challenge.

This comprehensive 20-page baseline standard project documentation provides an exhaustive breakdown of the technical, managerial, financial, and risk management parameters underpinning the WORKY platform. Developed under course **BS213: Project Management** at Innovation University's Faculty of Computers & Information Technology (FCIT), this document details the complete 80-day delivery lifecycle structured over 8 Agile Sprints with a total financial baseline cap of 35,120 EGP.

Core Value Proposition

WORKY combines real-time client-side GPS location services with an automated QR check-in engine and gamified loyalty mechanics. Users save time locating verified workspace environments, while co-working space owners increase venue occupancy and customer retention.

01. PROJECT CHARTER & GOVERNANCE

1.1 Project Overview & Purpose

The formal authorization of the WORKY App project is captured within this Project Charter. The fundamental mission of WORKY is to eliminate friction in discovering workspace infrastructure. Prior to WORKY, students and freelancers suffered productivity losses due to inaccurate online space listings, unexpected outlet failures, overbooked study lounges, and poor Wi-Fi bandwidth.

1.2 Governance Structure & Leadership

ROLE	ENTITY / INDIVIDUAL	PRIMARY AUTHORITY & RESPONSIBILITIES
Project Sponsor	Innovation University FCIT Board	Overall project approval, budgetary authorization, final academic sign-off.
Project Manager	Marwan Bahaa (ID: 24030006)	Schedule execution, resource allocation, risk management, and team governance.
Lead Developer	Mohamed Gouda (ID: 24030145)	Frontend/Backend architecture, PWA framework integration, Mapbox/GPS APIs.
QA & DB Architect	Mostafa Abdelkader (ID: 24030097)	Database schema enforcement, end-to-end testing, security & compliance audit.

1.3 Strategic Business Objectives

- **Minimize Search Friction:** Reduce average discovery and check-in time for study spaces from 25 minutes down to under 3 minutes.
- **Enhance Data Verification:** Guarantee 95%+ accuracy on listed amenities (power outlets, Wi-Fi speed, seating capacity) through crowdsourced and venue-verified badges.
- **Boost Partner Retention:** Increase repeat visits to participating venues by 30% through the integrated QR loyalty rewards subsystem.
- **Optimized Cost Structure:** Deliver a cross-platform solution fully operational within the 35,120 EGP financial cap over an 80-day execution window.

80 Days PROJECT TIMELINE	35,120 EGP BUDGET CAP	8 Sprints AGILE CADENCE	100% PWA CROSS-PLATFORM
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02. FEASIBILITY STUDY (TELOS FRAMEWORK)

To ensure long-term viability, project feasibility was rigorously evaluated across five strategic dimensions using the TELOS framework: Technical, Economic, Legal, Operational, and Schedule Feasibility.

2.1 Technical Feasibility (T)

WORKY utilizes a hybrid Progressive Web App (PWA) architecture utilizing React, TypeScript, and Node.js microservices hosted on Vercel and Firebase infrastructure. PWA technology guarantees full cross-platform compatibility across iOS and Android platforms without requiring separate native mobile codebases (Objective-C/Swift/Kotlin). This drastically reduces software development effort, eliminates App Store delay overhead, and enforces real-time client-side update capabilities.

2.2 Economic Feasibility (E)

The financial analysis demonstrates complete economic viability. The overall capital expenditure is capped at 35,120 EGP, which includes stipend allocations for development labor, cloud hosting tiers (Firebase DB, Google Maps/Mapbox APIs), design assets, and a 15% contingency reserve (6,000 EGP). Long-term monetization models include sponsored venue promotion placements and dynamic booking commission structures.

2.3 Legal & Regulatory Feasibility (L)

WORKY strictly adheres to Egyptian Cybercrime Law No. 175 and user data privacy directives. Geolocation tracking occurs exclusively on the client device upon explicit user opt-in, with zero persistent history tracking of personal trajectories. Database storage complies with AES-256 encryption at rest and TLS 1.3 transport security.

2.4 Operational Feasibility (O)

The user interfaces are tailored specifically for two distinct user demographics: students seeking intuitive map navigation, and venue administrators managing seats and validating QR vouchers. Onboarding for venue managers requires less than 15 minutes of training due to a simplified web admin portal.

2.5 Schedule Feasibility (S)

The project timeline is constrained to 80 calendar days, structured across 8 two-week continuous Agile Sprints. Critical path analysis confirms that key deliverables—from requirements spec to public deployment—can be executed with zero schedule slack, supported by a 5-day contingency buffer.

03. DETAILED PROJECT SCOPE STATEMENT

3.1 In-Scope Deliverables & Feature Modules

The project scope covers all research, design, engineering, testing, and deployment activities required to deliver the WORKY PWA platform.

MODULE REF	SCOPE DELIVERABLE	TECHNICAL & FUNCTIONAL DESCRIPTION
MOD-01	User Map & Discovery Engine	Real-time GPS positioning, dynamic Mapbox mapping, venue search, distance calculation, amenity filters (Wi-Fi, outlets, noise level).
MOD-02	Multi-Factor Authentication	Secure user registration via OAuth 2.0 (Google/Email), SMS OTP verification, and JWT session handling.
MOD-03	QR Check-in & Loyalty Subsystem	Dynamic QR generation, location verification match, point accumulation engine, and discount voucher redemption engine.
MOD-04	Venue Partner Dashboard	Admin panel for space owners to update real-time seat availability, verify student vouchers, and manage workspace amenity badges.
MOD-05	PWA Service Worker Engine	Offline caching of map tiles, background synchronization, push notifications, and desktop add-to-homescreen installation.

3.2 Out-of-Scope (Future Release Roadmap)

- Direct in-app payment gateway integration for credit/debit card seat reservations (Scheduled for Phase 2).
- Hardware kiosk integration or automated physical turnstile access control at co-working venues.
- Desktop-native offline executable software installations outside browser PWA capability.

04. WORK BREAKDOWN STRUCTURE (WBS)

The WBS decomposes the total project work into manageable deliverables and work packages across six main execution phases.

WBS CODE	PHASE / WORK PACKAGE	DETAILED ACTIVITY & DELIVERABLE BREAKDOWN
1.0	Project Initiation	1.1 Stakeholder Identification & Alignment 1.2 Draft & Sign Project Charter 1.3 Complete TELOS Feasibility Study
2.0	Requirements Analysis	2.1 User Story Harvesting & Persona Building 2.2 Functional & Non-Functional Specifications 2.3 GPS Precision Boundary & API Spec Mapping
3.0	UI/UX & Architecture	3.1 Figma Wireframing & Design System 3.2 React PWA Architecture Definition 3.3 Database Schema & API Route Design
4.0	Core Development	4.1 Map Engine & GPS Location Services 4.2 QR Code Engine & Points Subsystem 4.3 Partner Admin Portal Development
5.0	Testing & QA	5.1 Authentication & Multi-Factor Testing 5.2 GPS Latency & Precision Verification 5.3 Penetration Testing & Security Audit
6.0	Deployment & Launch	6.1 PWA Hosting & Vercel Configuration 6.2 User Onboarding & Marketing Launch 6.3 Final LMS Presentation Sign-Off

05. ACTIVITY LIST & SCHEDULE DEPENDENCIES

Each activity within the Work Breakdown Structure has been assigned precise durations, successor logic, and dependency linkages.

ID	ACTIVITY DESCRIPTION	PREDECESSOR	DURATION	RESOURCE ASSIGNED
ACT-01	Charter Approval & Stakeholder Sign-Off	None	5 Days	Marwan Bahaa
ACT-02	Requirements Specification & User Stories	ACT-01	10 Days	Marwan Bahaa / Mostafa Abdelkader
ACT-03	UI/UX Design & Component Design System	ACT-02	15 Days	Mohamed Gouda
ACT-04	Core PWA & GPS Map Engine Engine	ACT-03	25 Days	Mohamed Gouda / Mostafa Abdelkader
ACT-05	QR Code Loyalty & Check-in Subsystem	ACT-04	15 Days	Mohamed Gouda
ACT-06	Partner Admin Portal Development	ACT-04	10 Days	Mostafa Abdelkader
ACT-07	End-to-End System Testing & Security QA	ACT-05, ACT-06	15 Days	Mostafa Abdelkader / Marwan Bahaa
ACT-08	Production Deployment & Final Verification	ACT-07	10 Days	All Team Members

06. PROJECT COST ESTIMATION BASELINE

Cost estimation was conducted using a bottom-up methodology combined with analogous software development cost benchmarking tailored to the Egyptian market. Total budget cap is fixed at **35,120 EGP**.

6.1 Detailed Cost Breakdown

CATEGORY	RESOURCE / BREAKDOWN ITEM	COST (EGP)	BUDGET SHARE (%)
Development Labor	3 Software Engineers @ 8 Weeks (Stipend allocation)	21,000 EGP	59.8%
Cloud & APIs	Firebase Services, Vercel PWA hosting & Mapbox/Google Maps API tiers	5,120 EGP	14.6%
Design & Domain	UI/UX Asset licenses, Icon Sets, Custom Domain registration (.app)	3,000 EGP	8.5%
Contingency Reserve	15% Financial Risk Buffer (Unallocated for unexpected scope changes)	6,000 EGP	17.1%
TOTAL	APPROVED BASELINE BUDGET	35,120 EGP	100.0%

6.2 Cost Control & Management Rules

- Any expenditure exceeding 1,000 EGP from the contingency reserve requires formal approval from the Project Manager and Sponsor.
- Cloud API usage is rate-limited and monitored weekly to prevent threshold overruns.

07. EARNED VALUE MANAGEMENT (EVM) ANALYSIS

To evaluate performance and execution efficiency, Earned Value Management (EVM) metrics were calculated at the **Week 4 Midpoint Review** (Day 40).

7.1 Week 4 EVM Baseline Data Inputs

- **Planned Value (PV):** 17,560 EGP (50% of 35,120 EGP target)
- **Earned Value (EV):** 17,560 EGP (Work completed strictly aligns with planned scope)
- **Actual Cost (AC):** 17,000 EGP (Spend incurred to date)

7.2 EVM Performance Metrics Table

Metric Parameter	Mathematical Formula	Calculated Value	Status & Strategic Interpretation
Cost Variance (CV)	$EV - AC$	+560 EGP	Favorable Under budget by 560 EGP due to cloud tier optimization.
Schedule Variance (SV)	$EV - PV$	0 EGP	On Schedule Project execution precisely tracks planned timeline.
Cost Performance Index (CPI)	EV / AC	1.03	Favorable Generating 1.03 EGP of value for every 1.00 EGP spent.
Schedule Performance Index (SPI)	EV / PV	1.00	On Track Progress perfectly matches baseline schedule efficiency.

EVM Forecast at Completion

With $CPI = 1.03$, the Estimate at Completion (EAC) is calculated as $BAC / CPI = 35,120 / 1.03 = 34,097 \text{ EGP}$, projecting a net savings of 1,023 EGP upon project finalization.

08. PERT ANALYSIS & DURATION CALCULATIONS

Program Evaluation and Review Technique (PERT) was applied to quantify duration uncertainty across critical development work packages.

8.1 PERT Mathematical Formulas

Expected Duration: $T_e = (a + 4m + b) / 6$

Activity Variance: $\sigma^2 = [(b - a) / 6]^2$

8.2 Activity Duration Estimations Table

ACTIVITY DESCRIPTION	OPTIMISTIC (A)	MOST LIKELY (M)	PESSIMISTIC (B)	EXPECTED TIME (T _E)	VARIANCE (Σ ²)
Requirements & Spec Freeze	10 Days	7 Days	13 Days	10.0 Days	1.00
UI/UX Design Phase	12 Days	15 Days	18 Days	15.0 Days	1.00
PWA Core Development	20 Days	25 Days	30 Days	25.0 Days	2.78
Testing & Verification	10 Days	15 Days	20 Days	15.0 Days	2.78
Deployment & Launch	8 Days	10 Days	12 Days	10.0 Days	0.44

09. PROJECT SCHEDULE & GANTT TIMELINE (80 DAYS)

The project execution schedule spans exactly 80 calendar days, organized into 8 bi-weekly continuous Agile Sprints.

PROJECT PHASE / TASK	W1-W2	W3	W4-W5	W6-W7	W8
1. Requirements Analysis	REQ				
2. UI/UX & Schema Design		DESIGN			
3. Core PWA Development			DEV		
4. QA & GPS Testing				TEST	
5. Deployment & Launch					LAUNCH

9.1 Major Project Milestones

- **Milestone 1 (Day 15):** Requirements & Functional Specification Freeze.
- **Milestone 2 (Day 55):** PWA Core Engine & QR Module Beta Release.
- **Milestone 3 (Day 80):** Final Public Launch & LMS Academic Sign-Off.

10. CRITICAL PATH METHOD (CPM) & NETWORK DIAGRAM

Critical Path Analysis determines the sequence of dependent steps that defines the total minimum project duration.

10.1 CPM Activity Table

TASK NAME	DURATION	EARLY START (ES)	EARLY FINISH (EF)	LATE START (LS)	LATE FINISH (LF)	FLOAT / SLACK	CPM STATUS
Project Initiation	5 Days	0	5	0	5	0 Days	CRITICAL
Requirements Spec	10 Days	5	15	5	15	0 Days	CRITICAL
UI/UX Architecture	15 Days	15	30	15	30	0 Days	CRITICAL
PWA Development	25 Days	30	55	30	55	0 Days	CRITICAL
Testing & QA	15 Days	55	70	55	70	0 Days	CRITICAL
Deployment	10 Days	70	80	70	80	0 Days	CRITICAL

Critical Path Network Sequence

Path 1: Initiation → Requirements → UI/UX → Development → Testing → Deployment

Total Duration: 80 Days (Zero Slack). Any single day delay along this critical path directly delays project launch.

11. RISK MANAGEMENT & MITIGATION MATRIX

A proactive risk management framework was established to identify, analyze, and mitigate technical, operational, and financial risks.

RISK EVENT DESCRIPTION	CATEGORY	PROB	IMPACT	RISK LEVEL	MITIGATION STRATEGY & CONTINGENCY PLAN
GPS API Latency / Rate Limits Exceeded	Technical	Medium	High	HIGH	Implement client-side Mapbox caching and fallback geolocation positioning services.
Scope Creep (Unplanned Feature Requests)	Scope	High	Medium	MED-HIGH	Enforce strict Sprint Backlog freeze and require Change Control Board (CCB) approval.
Budget Overrun beyond baseline	Financial	Low	High	MEDIUM	Allocate 6,000 EGP dedicated financial contingency reserve buffer (15% cap).
Low Venue Partner Adoption Rate	Operational	Medium	Medium	MEDIUM	Offer zero-commission promotional tier for first 3 months to onboard initial venue partners.

12. TECHNICAL SOFTWARE ARCHITECTURE & TECHNOLOGY STACK

WORKY relies on a modular microservice PWA architecture optimized for rapid loading speed and low data consumption across mobile networks.

12.1 Technology Stack Selection

LAYER	TECHNOLOGY SELECTED	ENGINEERING JUSTIFICATION
Frontend Framework	React.js with TypeScript	Type safety, component reusability, modular architecture, fast virtual DOM rendering.
PWA Engine	Workbox / Service Workers	Enables offline tile caching, background data sync, push notifications, native installation.
Backend & Database	Node.js & Firebase / Firestore	Real-time database synchronization, serverless scalable function execution, quick OAuth.
Map Engine	Mapbox GL JS & Google Maps API	High precision custom vector maps, fast geocoding, client-side route calculation.

13. DETAILED MODULE: MAP ENGINE & GEOFENCING

The Map Engine forms the core user interaction surface of WORKY, allowing discovery of study hubs based on proximity and specific filters.

13.1 Key Capabilities

- **Real-time Location Detection:** HTML5 Geolocation API with high-accuracy GPS fallback.
- **Interactive Filtering:** Filter by verified Wi-Fi speed (>50 Mbps), power outlet density, quiet zone ratings, and seating availability.
- **Dynamic Distance Calculation:** Haversine mathematical algorithm implemented client-side for distance calculation.

14. DETAILED MODULE: QR LOYALTY SUBSYSTEM

To drive repeat student engagement and assist venues in verifying customer presence, WORKY incorporates an automated QR check-in engine.

14.1 QR Verification Workflow

1. User arrives at partner venue and opens WORKY PWA.
2. Venue displays dynamic, time-sensitive QR code on desk or portal.
3. User scans code via PWA camera integration.
4. System compares GPS coordinates of user device against venue location coordinates (Geofence radius: 50 meters).
5. Upon valid match, user receives instant loyalty points and voucher credits.

15. DETAILED MODULE: PARTNER BUSINESS PORTAL

Co-working space managers access a dedicated, simplified administrative interface to manage venue presence and track customer visits.

15.1 Core Admin Features

- **Live Occupancy Adjustment:** Toggle seat status (Available, Moderate, Full) in real-time.
- **Voucher Redemption Engine:** Scan student discount codes to prevent voucher fraud.
- **Analytics Dashboard:** View peak study hours, daily check-in volume, and customer retention metrics.

16. QUALITY ASSURANCE, SECURITY & COMPLIANCE

Rigorous testing and compliance protocols were enforced throughout software development to guarantee performance and security.

16.1 Security Controls

- **Data Encryption:** All sensitive payload transfers enforced over HTTPS / TLS 1.3 encryption.
- **Data Privacy:** Complete compliance with Egyptian Cybercrime Law No. 175. User location data is never logged persistently.
- **Authentication Security:** Secure JWT session management with automated expiration and refresh tokens.

17. STAKEHOLDER ENGAGEMENT & COMMUNICATION PLAN

Clear communication channels ensured seamless coordination between academic supervisors, the development team, and venue partners.

STAKEHOLDER GROUP	COMMUNICATION CHANNEL	FREQUENCY	PRIMARY OBJECTIVE
Academic Board / Instructors	Formal Sprint Reviews & Documentation	Bi-Weekly	Schedule and deliverable evaluation against LMS criteria.
Development Team	Daily Agile Stand-up & Slack	Daily (15 mins)	Task progress, impediment resolution, code reviews.
Partner Venues	Onsite Demonstrations & Feedback Forms	Weekly	Portal onboarding, QR validation testing, feature input.

18. LMS DELIVERABLES VERIFICATION & FACULTY SIGN-OFF

Confirmation of all 14 required LMS submission deliverables as mandated by Innovation University FCIT BS213 course guidelines.

ITEM #	MANDATED SUBMISSION DELIVERABLE	STATUS
1	GitHub Public Repository (Project2 Repository)	VERIFIED
2	Feasibility Study (TELOS Framework)	VERIFIED
3	Project Charter	VERIFIED
4	Project Scope Statement	VERIFIED
5	Work Breakdown Structure (WBS)	VERIFIED
6	Cost Estimation Baseline	VERIFIED
7	Earned Value Management (EVM) Metrics	VERIFIED
8	PERT Analysis & Duration Variance	VERIFIED
9	Gantt Timeline Chart (80 Days)	VERIFIED
10	Network Diagram Architecture	VERIFIED
11	Critical Path Method (CPM) Analysis	VERIFIED
12	Risk Management Matrix	VERIFIED
13	Final Project Presentation Deck	VERIFIED
14	Detailed Activity List	VERIFIED

FACULTY SIGN-OFF & ACADEMIC APPROVAL COMPLETE

Innovation University — Faculty of Computers & Information Technology (FCIT)
Course BS213: Project Management Board | All Baseline Standards Verified